Fully branch-wise DTL rates on the archaeal big tree: likelihood landscape, best fit, cross-validation, and a reproducible protocol

gpurec / AleRax comparison — This_study / Undine C60

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Abstract

We re-analyse the rooting of the archaeal "big tree" (This_study / Undine C60: 257 taxa, $S = 513$ species-tree branches, 7,059 gene families) with `gpurec`, a GPU re-implementation of AleRax's duplication–transfer–loss (DTL) gene-tree/species-tree reconciliation. Letting the DTL rates vary *freely along every branch* (∼2,000 parameters) recovers the same **Euryarchaeota** root and the same ancestral genome as AleRax, with duplication/transfer rates that are essentially clade-constant and loss rates that are branch-heterogeneous; held-out cross-validation shows this extra richness *generalises* rather than overfitting. The likelihood surface is multimodal: the global optimum (Euryarchaeota) is reached by warm-starting from a clade-grouped fit but missed by every cold start tried. We give the reproducible protocol, explain each analysis script, and state the open cold-start problem.

1 Background: what is being computed and why

A *gene family* is a set of homologous genes; its history inside the species tree is a sequence of events — the gene is *born* somewhere (*origination*, O), and then along each species branch it may be *duplicated* (D), *transferred* laterally to another branch (T), or *lost* (L). *Reconciliation* maps an observed (uncertain) gene tree onto the species tree and sums the probability over *all* such event scenarios, giving the marginal likelihood of that family under a set of D, L, T, O rates. Summing the log-likelihood over all 7,059 families gives the data log-likelihood we maximise.

The rooting question. The species tree's *root* is unknown. Each candidate root gives a different directed tree, hence a different likelihood; the maximum-likelihood root is the inferred answer. For archaea the contested answer is whether the root falls on **Euryarchaeota** (the result we are testing) or is pulled onto small-genome lineages by a known artifact (*small-genome attraction*, SGA).

Clade-grouped vs. branch-wise rates. AleRax does not give every branch its own rates; it groups branches into a handful of rate *classes* (e.g. one D, L, T per major clade) to keep the model parsimonious. The question of this report is what happens when we *remove* that constraint and let every one of the 513 branches carry its own D, L, T (and origination weight) — a "fully branch-wise" model with ∼2,000 free parameters. Does it still give Euryarchaeota? Does the extra freedom just fit noise?

gpurec. `gpurec` is a GPU (PyTorch + Triton) re-implementation of AleRax's DTL reconciliation. It computes the same marginal likelihood and its gradient with respect to the rates (by implicit differentiation), so the rates can be optimised. Evaluated at AleRax's exact rates it reproduces AleRax's per-family log-likelihoods to **Pearson** $r = 0.999998$ for every candidate root and model class — so any disagreement below is about the *optimiser* or the *model richness*, never the engine.

Throughout: rates are reported as natural (per-branch) rates; log-likelihoods are in *nats* (natural log); fraction-missing data is handled in the AleRax-faithful `e-only` mode.

2 The likelihood landscape: three basins

Maximising the likelihood does not give a single answer: the surface is *multi-modal*, with several local optima ("basins"). Which one the optimiser reaches depends on where it starts. The coordinate that distinguishes the basins is the *origination* distribution O — where on the tree gene families are born:

- a **deep, concentrated** O (births clustered near the Euryarchaeota/root region) $\rightarrow$ the Euryarchaeota root;
- a **diffuse** O (births spread thinly everywhere) $\rightarrow$ a near-tie among roots, or the SGA artifact.

O is therefore the rooting signal, and it is *bistable*. Table 1 lists the three basins we observe.

Why a cold start fails. The deep basin is the highest likelihood, but its "catchment" on the parameter surface is small: from a uniform (uninformed) start the optimiser slides into the much larger near-tie or SGA basins and never climbs out. Reaching the deep basin requires the origination O *and* the DTL rates to be jointly near the good solution — they are coupled — which is why a joint warm-start works and partial nudges do not (§7).

Basin	data $\log L$ (nats)	how it is reached
Deep / Euryarchaeota (global best)	$-1{,}631{,}537$	warm-start from a clade-grouped fit
Cold near-tie	$-1{,}664{,}248$ (Eury)	cold global-rate warm-up
Cold SGA stall	$\approx -1{,}712{,}000$	cold uniform initialisation

Table 1: The three basins. The deep/Euryarchaeota basin is the global optimum found, $\sim$33,000 nats above the cold near-tie and $\sim$80,000 above the cold SGA stall. In the cold near-tie basin the SGA root Cluster2 ($-1{,}663{,}618$) marginally *edges* Eury ($-1{,}664{,}248$); only in the deep basin does Eury win decisively. The initialisation selects the basin.

3 The best fit: fully branch-wise ("FULLbasin")

The best fit lets every branch have its own D, L, T and origination weight ($\sim$2,000 parameters), warm-started from the clade-grouped fit so that it sits in the deep basin. It reaches $\log L = -1{,}631{,}537$ nats, the highest found and $\sim$11,000 nats above the clade-grouped fit. Its properties:

Rooting. Euryarchaeota is rank 1 by $+3{,}568$ nats over the next root; the deep clades {Eury, TackA, DPANN} sit at the top and the small-genome (SGA) roots at the bottom (Cluster2 is $-12{,}190$ nats below Eury). So relaxing the clade constraint does not change the root.

AU test — how confident is the root? An approximately-unbiased (AU) test (Shimodaira multiscale bootstrap) asks which roots are *statistically indistinguishable* from the best, given the scatter of per-family likelihoods — the set it returns is the rooting confidence set. On the free per-branch likelihoods it returns **Euryarchaeota alone** (BP= 1.0); every other root, *including MHH* ($-1{,}637{,}704$, rank 4), is rejected. AleRax's parsimonious clade model instead returns a *region* {Euryarchaeota, MHH} (MHH only 9.7 nats behind). So the free model is much *sharper*. We read this as a consequence of model richness plus the warm-start, not as stronger evidence — and we test directly whether it is overfitting in §4.

Rate structure (Fig. 1). A striking pattern emerges when the rates are free. Duplication and transfer come out essentially *clade-constant*: clade membership (the paper's 23 DTL classes) explains $R^2 \approx 0.94$ of their branch-to-branch variance, so the per-branch estimates barely depart from one rate per clade. Loss is the opposite — only $R^2 \approx 0.38$ of its variance is clade-level; it varies substantially *within* clades. Most of the free model's $\sim$11,000-nat advantage lives in these loss rates, i.e. the data genuinely call for finer-grained loss variation than a clade model allows.

Origination. The fitted O is concentrated but free: two branches carry $\sim$32% of the origination mass (top branch 0.19), over a near-uniform diffuse floor (73 branches hold 50%, 279 hold 90% of the mass) — the "concentrated-deep plus background" shape that distinguishes the deep basin.

Ancestral genome / LACA (Fig. 2). Counting the families inferred to be present at the root (posterior ≥ 0.5) gives the reconstructed last archaeal common ancestor genome. The free model places **903** families at the root, the clade-grouped (AleRax) rates **772**; **722** are shared (Jaccard 0.76), and the per-family origination-at-root probabilities correlate $r = 0.96$. So the two models agree on largely the *same* ancestral gene set, with the free model reconstructing a somewhat larger genome.

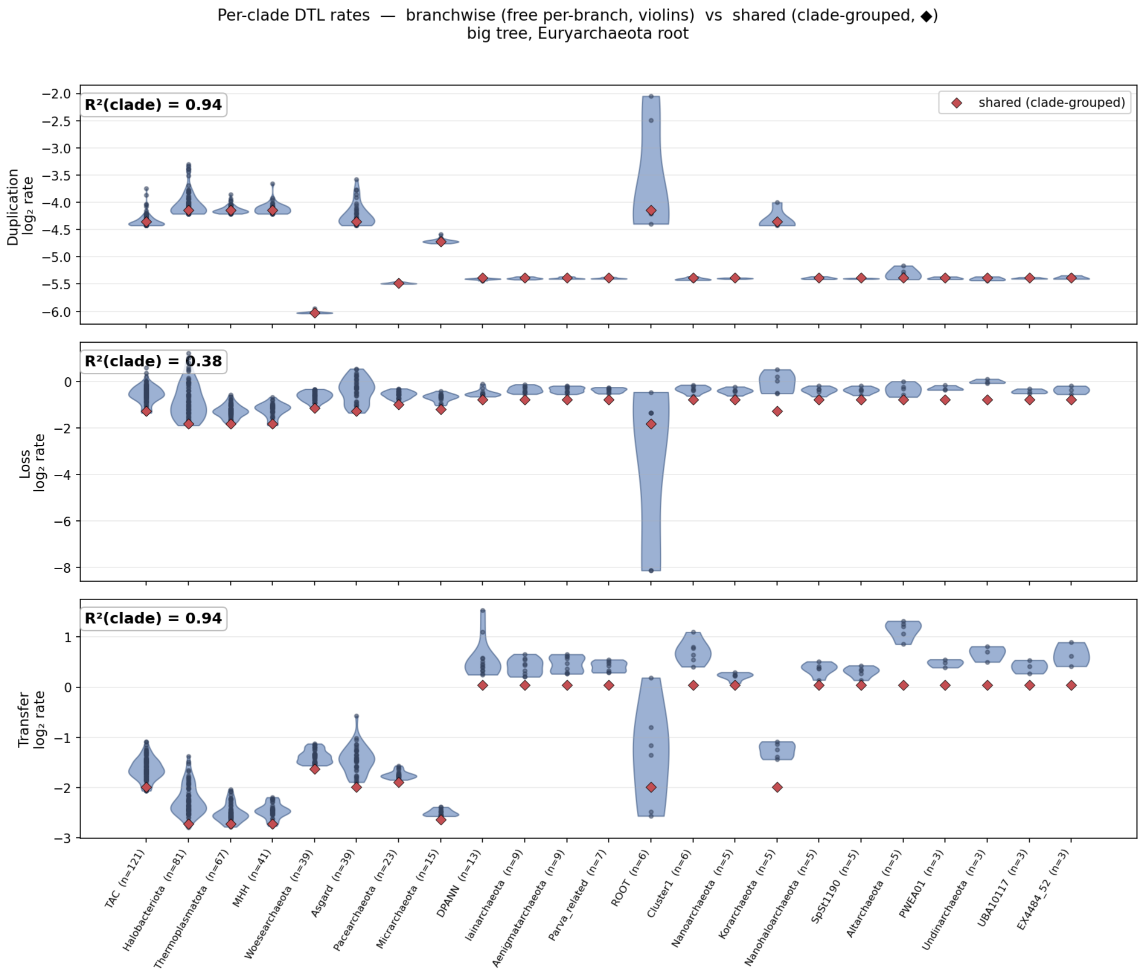

Figure 1: Per-clade DTL rates: *branchwise* (free per-branch, blue violins = the spread of branch rates within a clade) vs. *shared* (clade-grouped, red ⧫), one panel per rate. Duplication and transfer violins are tight around the shared rate ($R^2 \approx 0.94$ — clade-constant); loss violins are wide ($R^2 \approx 0.38$ — heterogeneous within clades).

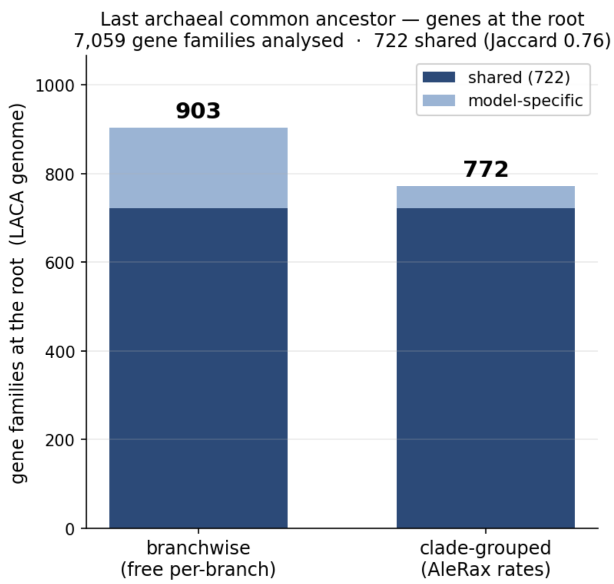

Figure 2: Gene families at the root (LACA): 903 under the free per-branch model vs. 772 under clade-grouped (AleRax) rates; 722 shared (dark, Jaccard 0.76), of 7,059 families analysed.

4 Cross-validation: is the extra richness overfitting?

A model with ∼2,000 free parameters could be fitting noise. The standard test is *out-of-sample prediction*: fit the rates on one set of families, then score *different* families that were never used in fitting. Overfit parameters help on the training families but *hurt* on held-out ones; genuinely useful parameters help on both. We fit both models on a training half (first 3,530 families) and scored the held-out half (3,529 families) — Table 2.

Scope of the claim. This proves the per-branch *rate* richness generalises (the core of the overfitting worry). It does not separately cross-validate the AU *region* (which would need re-fitting every root on the training set). The index split is also family-size unbalanced (training −389/fam vs. test −73/fam — the first half holds the larger families), so this is in fact a harder test, *across* a family-size shift, which it passes; a random split would give a cleaner balanced number.

model	train $\sum \log L$	held-out $\sum \log L$	train/fam	held-out/fam
free (per-branch)	−1,373,618	−257,725	−389.13	−73.03
grouped (clade)	−1,383,170	−259,280	−391.83	−73.47
free − grouped	+9,551	**+1,556**		+0.44

Table 2: Held-out cross-validation (Eury root). On families never used in fitting, the free per-branch model still beats clade-grouped by +1,556 nats (+0.44/family). The advantage shrinks from +9,551 (train) to +1,556 (held-out) — about five-sixths of the in-sample gain was optimism, but the remaining positive sixth *generalises*. **Verdict: not overfitting.**

5 Reproducible protocol (the minimal robust setup)

The fitting steps use the general `gpurec fit` command, which takes the inputs *explicitly* (`--species-tree`, `--ale-dir`) rather than any dataset-specific naming convention — so the same commands apply to any species tree + gene-family set. Run from a `gpurec` checkout on a single A100 (`largegpu`); `gpurec fit --help` lists every flag. The shell variables below name the output directory (`$U`), the data root (`$DD`), the species tree (`$SP`), the gene-family directory (`$ALE`), and the AleRax clade-grouping directory (`$CG`).

```
U=/work/SzollosiU/gergely-szollosi/williams_run/undine
DD=/work/SzollosiU/gergely-szollosi/williams_run/data/3_Reconciliation/This_study
SP=$DD/4_species_tree/Undine_C60_Euryroot_short_name.nw # explicit rooted species tree
ALE=$DD/3_UFBOOTs/ufboot_for_alerax # directory of .ale CCPs (7059)
FM=$DD/fraction_missing # fraction_missing file
CG="$U/alerax_ref/3_Reconciliation/This_study/5_reconcilation␣models/DTL_br1_O/
    ↪ Undine_C60_Euryroot2_OR"
```

Step 1 — clade-grouped warm-up (selects the deep basin, ∼30 min). Fit the clade-grouped DTL+O model, seeded from AleRax's rates. This is the step that lands in the deep/Euryarchaeota basin.

```
gpurec fit --species-tree "$SP" --ale-dir "$ALE" --fraction-missing "$FM" \
  --min-species 1 --fm-mode e-only --prior none \
  --clade-groups "$CG" --init-from-alerax --origination optimize \
  --family-batch-size 500 --steps 250 --dtype float64 \
  --out $U/AXgrp_DTL_br1_O_Eury.rates.txt
```

Step 2 — fully free per-branch fit (the best fit, ∼1–7.5 h). Warm-started from Step 1; no rate grouping, free per-branch origination.

```
gpurec fit --species-tree "$SP" --ale-dir "$ALE" --fraction-missing "$FM" \
  --min-species 1 --fm-mode e-only --prior none --origination optimize \
  --init-from-rates $U/AXgrp_DTL_br1_O_Eury.rates.txt.json \
  --family-batch-size 500 --steps 250 --dtype float64 \
  --out $U/FULLbasin_Eury.rates.txt
```

Step 3 — rooting / AU. Run Steps 1–2 for every root (point `--species-tree` at that root's tree and `--clade-groups` at its matching grouping dir), then:

```
python experiments/au_fullbasin.py # AU test over the 15 per-root fits
```

Step 4 — ancestral genome (LACA).

```
python experiments/laca_compare.py --root Eury \
  --sidecar $U/FULLbasin_Eury.rates.txt.json --out $U/laca_Eury.json
```

Step 5 — overfitting cross-validation. Fit both models on the training half (`--families 3530`), then score held-out:

```
gpurec fit --species-tree "$SP" --ale-dir "$ALE" --fraction-missing "$FM" \
  --families 3530 --min-species 1 --fm-mode e-only --prior none --clade-groups "$CG" \
  --init-from-alerax --origination optimize --family-batch-size 500 \
  --steps 250 --dtype float64 --out $U/CVgrouped_Eury_train.rates.txt
gpurec fit --species-tree "$SP" --ale-dir "$ALE" --fraction-missing "$FM" \
  --families 3530 --min-species 1 --fm-mode e-only --prior none --origination optimize \
  --init-from-rates $U/CVgrouped_Eury_train.rates.txt.json \
  --family-batch-size 500 --steps 250 --dtype float64 \
  --out $U/CVfree_Eury_train.rates.txt
python experiments/cv_eval.py --root Eury --train-n 3530 \
  --free $U/CVfree_Eury_train.rates.txt.json \
  --grouped $U/CVgrouped_Eury_train.rates.txt.json --out $U/cv_eval_Eury.json
```

Key flags. `--fm-mode e-only` (AleRax-faithful fraction-missing); `--prior none` (no smoothing; `--dtl-tv-lambda` adds a fused-lasso that auto-groups the free rates); `--origination optimize` (fit per-branch origination — the rooting signal); `--clade-groups <dir>` (use AleRax's clade rate-classes) with `--init-from-alerax` (seed from AleRax's rates); `--init-from-rates <sidecar>` (warm-start chain); `--family-batch-size 500` (the S=513, 7059-family forward must be batched or it overflows the GPU kernels); `--dtype float64` (gradient accuracy); `--families N` (use the first N families; 0 = all).

6 What each script does

The protocol uses a small number of scripts under `experiments/`. Their roles:

`gpurec fit` **— the fitter (general CLI).** The workhorse, exposed as a general, dataset-agnostic command: `gpurec fit --species-tree <Newick> --ale-dir <dir> ...` (no dataset-specific naming convention; `gpurec fit --help` lists every flag). It loads the rooted species tree and amalgamates the gene-family conditional-clade distributions from the `.ale` files (the C++ pre-processing), lays the families out in cross-family "waves" for the GPU, and then *maximises* the total reconciliation log-likelihood over the DTL(+O) rates by L-BFGS. Each optimisation step evaluates the likelihood by two nested fixed points (extinction E per branch, then the family likelihoods Π) and gets the gradient by implicit differentiation. Flags choose the model (free per-branch, or `--clade-groups` for AleRax's classes), the initialisation (`--init-from-alerax`, `--init-from-rates`), and any prior (`--prior`, `--dtl-tv-lambda`, `--origination-root-lambda`). It writes a `.rates.txt` and a `.rates.txt.json` "sidecar" holding the fitted $\log_2$ per-branch rates (`theta_log2`), the origination weights, the data log-likelihood, and the exact command used — every downstream script reads this sidecar. (Implementation: `gpurec/cli/fit.py` over the validated branch-wise driver `experiments/run_undine_branchwise.py`.)

`au_fullbasin.py` — **rooting confidence set.** Reads the per-root FULLbasin sidecars, recomputes every family's log-likelihood under each root, and runs the Shimodaira AU test (multiscale bootstrap) to return the set of roots that cannot be statistically rejected. This is what yields "Euryarchaeota alone" (§3).

`laca_compare.py` — **ancestral genome.** At a chosen root, computes each family's posterior probability of *originating at the root* under two rate sets (the free per-branch fit and AleRax's rates), thresholds at 0.5 to get the LACA gene set for each, and reports the two genome sizes, their overlap (Jaccard), and the per-family probability correlation. (AleRax stored only per-family likelihoods, no reconciliation, so the "AleRax" side is `gpurec` evaluated at AleRax's exact rates — a faithful proxy, given $r = 0.999998$.) Produces Fig. 2.

`cv_eval.py` — **overfitting verdict.** Given two sidecars fit on the training half (free and grouped), it computes the per-family log-likelihood for *all* families and splits them into the training and held-out halves, reporting each model's train and held-out totals. Free > grouped on the held-out half is the not-overfitting verdict (Table 2).

`rate_violin_data.py` / `plot_rate_violin.py` — **rate-structure figure.** The first dumps, for every branch, its free per-branch rate, its clade-grouped (shared) rate, and its clade label; the second draws the per-clade violins of branch rates with the shared rate overlaid (Fig. 1).

`plot_laca_families.py` — **LACA bar chart.** Draws Fig. 2 from `laca_compare.py`'s output.

7 Open problem: cold-start basin-finding

The deep/Euryarchaeota basin is the global optimum but is only reached by warm-starting (Steps 1–2). From a cold (uniform-origination) start, every strategy tried so far lands in the near-tie or SGA basin:

- anti-concentration (Simpson) origination barrier — *directionally wrong*: it pushes O toward uniform, which sinks Euryarchaeota;
- fixed root-mass or depth origination priors — over-constrain and destroy the fit;
- global-rate warm-up — reaches only the near-tie ($-1{,}664$k);
- fused-lasso (TV) annealing — regularises the DTL, not the (bistable) origination, and sticks at the near-tie;
- structured-O seed ($\omega \propto -\text{depth}$) — diffuses straight back to the near-tie.

The deep basin needs the structured O *and* the matching DTL together; an initialisation-independent route to it is unsolved. The most promising untried idea — now running — is a *deep-origination prior homotopy*: impose a strong root-region origination prior to *place* the optimiser in the deep basin, then anneal that prior to zero, so the final unconstrained optimum is the one near the deep basin. This is the directionally-correct counterpart of the anti-concentration barrier that failed.